

Assignment-6.3

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Task Description #1 (Loops – Automorphic Numbers in a Range) •

Task: Prompt AI to generate a function that displays all Automorphic numbers between 1 and 1000 using a for loop.

• Instructions:

- o Get AI-generated code to list Automorphic numbers using a for loop.
- o Analyze the correctness and efficiency of the generated logic.
- o Ask AI to regenerate using a while loop and compare both implementations.

Expected Output #1:

- Correct implementation that lists Automorphic numbers using both loop types, with explanation.

Code:

```
C:\Users > PC > Downloads > aiac > assign_6.py > ...
1 # generate a python code to display all automorphic numbers between the range 1 to 1000 using for loop
2 import time as t
3 def is_automorphic(num):
4     square = num * num
5     return str(square).endswith(str(num))
6 def automorphic_numbers_in_range(start, end):
7     automorphic_numbers = []
8     for num in range(start, end + 1):
9         if is_automorphic(num):
10             automorphic_numbers.append(num)
11     return automorphic_numbers
12 start_time = t.time()
13 start_range = 1
14 end_range = 1000
15 automorphic_numbers = automorphic_numbers_in_range(start_range, end_range)
16 print(f"Automorphic numbers between {start_range} and {end_range} are: {automorphic_numbers}")
17 end_time = t.time()
18 print(f"Execution time: {end_time - start_time} seconds")
```

```
# generate a python code to display all automorphic numbers between the range 1 to 1000 using while loop
import time as t
def is_automorphic(num):
    square = num * num
    return str(square).endswith(str(num))
def automorphic_numbers_in_range(start, end):
    automorphic_numbers = []
    num = start
    while num <= end:
        if is_automorphic(num):
            automorphic_numbers.append(num)
        num += 1
    return automorphic_numbers
start_time = t.time()
start_range = 1
end_range = 1000
automorphic_numbers = automorphic_numbers_in_range(start_range, end_range)
print(f"Automorphic numbers between {start_range} and {end_range} are: {automorphic_numbers}")
end_time = t.time()
print(f"Execution time: {end_time - start_time} seconds")
```

Output:

```
PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe c:/Users/PC/Downloads/aiac/assign_6.py
Automorphic numbers between 1 and 1000 are: [1, 5, 6, 25, 76, 376, 625]
Execution time: 0.0007140636444091797 seconds
Automorphic numbers between 1 and 1000 are: [1, 5, 6, 25, 76, 376, 625]
Execution time: 0.0006761550903320312 seconds
PS C:\Users\PC>
```

Explanation : The for loop is more Pythonic and cleaner than the manual while loop. Prefer for loops when iterating over a known range. Both implementations produce identical results and similar execution times.

Task Description #2 (Conditional Statements – Online Shopping Feedback Classification)

- Task: Ask AI to write nested if-elif-else conditions to classify online shopping feedback as Positive, Neutral, or Negative based on a numerical rating (1–5).

- Instructions:

- o Generate initial code using nested if-elif-else. o Analyze correctness and readability.

- o Ask AI to rewrite using dictionary-based or match-case structure.

Expected Output #2:

- Feedback classification function with explanation and an alternative Approach

CODE :

```
C:\Users\> PC> Downloads> aia> assign_6(2).py >
1 #generate a python code to get online shopping feedback classification using nested if else conditions as positive,negative or neutral based on numerical rating from 1 to
2 def classify_feedback(rating):
3     if rating >= 4:
4         return "Positive"
5     else:
6         if rating == 3:
7             return "Neutral"
8         else:
9             return "Negative"
10
11 # Example usage
12 rating = int(input("Enter your rating (1-5): "))
13 feedback = classify_feedback(rating)
14 print(f"Feedback classification: {feedback}")
15
16 # rewrite the above code using dictionaries
17 def classify_feedback_dict(rating):
18     feedback_dict = {
19         5: "Positive",
20         4: "Positive",
21         3: "Neutral",
22         2: "Negative",
23         1: "Negative"
24     }
25     return feedback_dict.get(rating, "Invalid rating")
26
27 # Example usage
28 rating = int(input("Enter your rating (1-5): "))
29 feedback = classify_feedback_dict(rating)
30 print(f"Feedback classification: {feedback}")
```

OUTPUT :

```
PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aia/assign_6(2).py"
Enter your rating (1-5): 3
Feedback classification: Neutral
Enter your rating (1-5): 4
Feedback classification: Positive
```

Explanation :

The dictionary approach is more Pythonic and preferred for mapping-style problems. Both produce identical output for valid inputs (1-5), but the dictionary version handles edge cases better with its default parameter.

Task 3: Statistical_operations

Define a function named statistical_operations(tuple_num) that performs the following statistical operations on a tuple of numbers:

- Minimum, Maximum
- Mean, Median, Mode
- Variance, Standard Deviation

While writing the function, observe the code suggestions provided by GitHub Copilot. Make decisions to accept, reject, or modify the suggestions based on their relevance and correctness.

Code :

```

C: > Users > PC > Downloads > aiac > assign_6(3).py > statistical_operations
1 def statistical_operations(data):
2     """
3     Perform basic statistical operations on a list of numbers.
4
5     Parameters:
6     data (list of float): A list of numerical values.
7
8     Returns:
9     dict: A dictionary containing the mean, median, and standard deviation.
10    """
11    import statistics
12
13    if not data:
14        return {"mean": None, "median": None, "std_dev": None}
15
16    mean = statistics.mean(data)
17    median = statistics.median(data)
18    std_dev = statistics.stdev(data) if len(data) > 1 else 0.0
19
20    return {
21        "mean": mean,
22        "median": median,
23        "std_dev": std_dev
24    }
25
26    # Example usage
27    data = [10, 20, 30, 40, 50]
28    results = statistical_operations(data)
29    print("Statistical Operations Results:")
30    print(f"Mean: {results['mean']}")
31    print(f"Median: {results['median']}")
32    print(f"Standard Deviation: {results['std_dev']}")

```

Output :

```

PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aiac/assign_6(3).py"
Statistical Operations Results:
Mean: 30
Median: 30
Standard Deviation: 15.811388300841896

```

Task 4: Teacher Profile

- Prompt: Create a class Teacher with attributes teacher_id, name, subject, and experience. Add a method to display teacher details.
- Expected Output: Class with initializer, method, and object creation.

Code :

```

C: > Users > PC > Downloads > aiac > assign_6(4).py > ...
1 class Teacher:
2     def __init__(self, name, id, subject, experience):
3         self.name = name
4         self.id = id
5         self.subject = subject
6         self.experience = experience
7
8     def display_info(self):
9         print(f"Teacher Name: {self.name}")
10        print(f"Teacher ID: {self.id}")
11        print(f"Subject: {self.subject}")
12        print(f"Years of Experience: {self.experience}")
13
14    # Example usage
15    teacher1 = Teacher("Alice Johnson", "T123", "Mathematics", 10)
16    teacher1.display_info()

```

Output :

```

PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aiac/assign_6(4).py"
Teacher Name: Alice Johnson
Teacher ID: T123
Subject: Mathematics
Years of Experience: 10

```

Task #5 – Zero-Shot Prompting with Conditional Validation

Use zero-shot prompting to instruct an AI tool to generate a function that validates an Indian mobile number.

Requirements

- The function must ensure the mobile number:
 - o Starts with 6, 7, 8, or 9
 - o Contains exactly 10 digits

Expected Output

- A valid Python function that performs all required validations without using any input-output examples in the prompt.

Code :

```

C: > Users > PC > Downloads > aiac > assign_6(5).py > ...
1 #generate a python code that validates an Indian mobile number.
2 import re
3
4 def validate_indian_mobile_number(mobile_number):
5     """
6     Validates an Indian mobile number.
7
8     Parameters:
9     mobile_number (str): The mobile number to be validated.
10
11     Returns:
12     bool: True if the mobile number is valid, False otherwise.
13     """
14     # Indian mobile numbers are 10 digits long and start with 6, 7, 8, or
15     pattern = r'^[6-9]\d{9}$'
16
17     if re.match(pattern, mobile_number):
18         return True
19     else:
20         return False
21
22 # Example usage
23 mobile_number = "9876543210"
24 if validate_indian_mobile_number(mobile_number):
25     print(f"{mobile_number} is a valid Indian mobile number.")
26 else:
27     print(f"{mobile_number} is not a valid Indian mobile number.")

```

Output :

```

PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aiac/assign_6(5).py"
9876543210 is a valid Indian mobile number.

```

Task Description #6 (Loops – Armstrong Numbers in a Range)

Task: Write a function using AI that finds all Armstrong numbers in a user-specified range (e.g., 1 to 1000).

Instructions:

- Use a for loop and digit power logic.
- Validate correctness by checking known Armstrong numbers (153, 370, etc.).
- Ask AI to regenerate an optimized version (using list comprehensions).

Expected Output #7:

- Python program listing Armstrong numbers in the range.
- Optimized version with explanation.

Code:

```

C: > Users > PC > Downloads > aiac > assign_6(6).py > ...
1 # generate a python code to display the armstrong numbers in a given range using loops
2 start = int(input("Enter the start of the range: "))
3 end = int(input("Enter the end of the range: "))
4
5 def is_armstrong(num):
6     num_str = str(num)
7     num_digits = len(num_str)
8     sum_of_powers = 0
9     for digit in num_str:
10         sum_of_powers += int(digit) ** num_digits
11     return sum_of_powers == num
12
13 print(f"Armstrong numbers in the range {start} to {end}:")
14 for i in range(start, end + 1):
15     if is_armstrong(i):
16         print(i)

```

Output :

```

PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aiac/assign_6(6).py"
Enter the start of the range: 1
Enter the end of the range: 1000
Armstrong numbers in the range 1 to 1000:
1
2
3
4
5
6
7
8
9
153
370
371
407
PS C:\Users\PC>

```

Task Description #7 (Loops – Happy Numbers in a Range)

Task: Generate a function using AI that displays all Happy Numbers within a user-specified range (e.g., 1 to 500).

Instructions:

- Implement the logic using a loop: repeatedly replace a number with the sum of the squares of its digits until the result is either 1 (Happy Number) or enters a cycle (Not Happy).
- Validate correctness by checking known Happy Numbers (e.g., 1, 7, 10, 13, 19, 23, 28...).
- Ask AI to regenerate an optimized version (e.g., by using a set to detect cycles instead of infinite loops).

Expected Output #8:

- Python program that prints all Happy Numbers within a range.
- Optimized version using cycle detection with explanation.

Code:

```
C:\> Users > PC > Downloads > aiaa > assign_6(7).py > ...
1  # generate a python code to display all happy numbers within the range
2  def is_happy_number(n):
3      seen = set()
4      while n != 1 and n not in seen:
5          seen.add(n)
6          n = sum(int(digit) ** 2 for digit in str(n))
7      return n == 1
8  def happy_numbers_in_range(start, end):
9      happy_numbers = []
10     for num in range(start, end + 1):
11         if is_happy_number(num):
12             happy_numbers.append(num)
13     return happy_numbers
14  if __name__ == "__main__":
15     start = int(input("Enter the start of the range: "))
16     end = int(input("Enter the end of the range: "))
17     happy_numbers = happy_numbers_in_range(start, end)
18     print(f"Happy numbers between {start} and {end}: {happy_numbers}")
19
```

```
#optimize the above code with cycle detection
def is_happy_number(n):
    def get_next(number):
        return sum(int(digit) ** 2 for digit in str(number))

    slow = n
    fast = get_next(n)

    while fast != 1 and slow != fast:
        slow = get_next(slow)
        fast = get_next(get_next(fast))

    return fast == 1
def happy_numbers_in_range(start, end):
    happy_numbers = []
    for num in range(start, end + 1):
        if is_happy_number(num):
            happy_numbers.append(num)
    return happy_numbers
if __name__ == "__main__":
    start = int(input("Enter the start of the range: "))
    end = int(input("Enter the end of the range: "))
    happy_numbers = happy_numbers_in_range(start, end)
    print(f"Happy numbers between {start} and {end}: {happy_numbers}")
```

Output :


```

PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aia/assign_6(7).py"
Enter the start of the range: 1
Enter the end of the range: 1000
Happy numbers between 1 and 1000: [1, 7, 10, 13, 19, 23, 28, 31, 32, 44, 49, 68, 70, 79, 82, 86, 91, 94, 97, 100, 103, 109, 129, 130, 133, 139, 167, 176, 188,
190, 192, 193, 203, 208, 219, 226, 230, 236, 239, 262, 263, 280, 291, 293, 301, 302, 310, 313, 319, 320, 326, 329, 331, 338, 356, 362, 365, 367, 368, 376, 37
9, 383, 386, 391, 392, 397, 404, 409, 440, 446, 464, 469, 478, 487, 490, 496, 536, 556, 563, 565, 566, 608, 617, 622, 623, 632, 635, 637, 638, 644, 649, 653,
655, 656, 665, 671, 673, 680, 683, 694, 700, 709, 716, 736, 739, 748, 761, 763, 784, 790, 793, 802, 806, 818, 820, 833, 836, 847, 860, 863, 874, 881, 888, 899
, 901, 904, 907, 910, 912, 913, 921, 923, 931, 932, 937, 940, 946, 964, 970, 973, 989, 998, 1000]
Enter the start of the range: 1
Enter the end of the range: 1000
Happy numbers between 1 and 1000: [1, 7, 10, 13, 19, 23, 28, 31, 32, 44, 49, 68, 70, 79, 82, 86, 91, 94, 97, 100, 103, 109, 129, 130, 133, 139, 167, 176, 188,
190, 192, 193, 203, 208, 219, 226, 230, 236, 239, 262, 263, 280, 291, 293, 301, 302, 310, 313, 319, 320, 326, 329, 331, 338, 356, 362, 365, 367, 368, 376, 37
9, 383, 386, 391, 392, 397, 404, 409, 440, 446, 464, 469, 478, 487, 490, 496, 536, 556, 563, 565, 566, 608, 617, 622, 623, 632, 635, 637, 638, 644, 649, 653,
655, 656, 665, 671, 673, 680, 683, 694, 700, 709, 716, 736, 739, 748, 761, 763, 784, 790, 793, 802, 806, 818, 820, 833, 836, 847, 860, 863, 874, 881, 888, 899
, 901, 904, 907, 910, 912, 913, 921, 923, 931, 932, 937, 940, 946, 964, 970, 973, 989, 998, 1000]
PS C:\Users\PC>

```

Explanation :

The first approach uses $O(k)$ space where k is the number of unique values encountered before determining happiness. The second approach uses $O(1)$ constant space but performs more function calls. For typical ranges, both are fast—the optimization shines with very large numbers or massive ranges where memory becomes a bottleneck.

Task Description #8 (Loops – Strong Numbers in a Range)

Task: Generate a function using AI that displays all Strong Numbers (sum of factorial of digits equals the number, e.g., $145 = 1! + 4! + 5!$) within a given range.

Instructions:

- Use loops to extract digits and calculate factorials.
- Validate with examples (1, 2, 145).
- Ask AI to regenerate an optimized version (precompute digit factorials).

Expected Output #9:

- Python program that lists Strong Numbers.
- Optimized version with explanation.

Code :

```

1 #generate a python code to display all the strong numbers within a given range using loops
2 def factorial(n):
3     if n == 0 or n == 1:
4         return 1
5     fact = 1
6     for i in range(2, n + 1):
7         fact *= i
8     return fact
9
10 def is_strong_number(num):
11     sum_of_factorials = 0
12     temp = num
13     while temp > 0:
14         digit = temp % 10
15         sum_of_factorials += factorial(digit)
16         temp //= 10
17     return sum_of_factorials == num
18
19 def find_strong_numbers_in_range(start, end):
20     strong_numbers = []
21     for num in range(start, end + 1):
22         if is_strong_number(num):
23             strong_numbers.append(num)
24     return strong_numbers
25
26 if __name__ == "__main__":
27     start_range = int(input("Enter the start of the range: "))
28     end_range = int(input("Enter the end of the range: "))
29     strong_numbers = find_strong_numbers_in_range(start_range, end_range)
30     print(f"Strong numbers between {start_range} and {end_range} are: {strong_numbers}")

```

```

#optimize the above code
def factorial(n, memo={}):
    if n in memo:
        return memo[n]
    if n == 0 or n == 1:
        return 1
    fact = 1
    for i in range(2, n + 1):
        fact *= i
    memo[n] = fact
    return fact
def is_strong_number(num):
    sum_of_factorials = 0
    temp = num
    while temp > 0:
        digit = temp % 10
        sum_of_factorials += factorial(digit)
        temp //= 10
    return sum_of_factorials == num
def find_strong_numbers_in_range(start, end):
    strong_numbers = []
    for num in range(start, end + 1):
        if is_strong_number(num):
            strong_numbers.append(num)
    return strong_numbers
if __name__ == "__main__":
    start_range = int(input("Enter the start of the range: "))
    end_range = int(input("Enter the end of the range: "))
    strong_numbers = find_strong_numbers_in_range(start_range, end_range)
    print(f"Strong numbers between {start_range} and {end_range} are: {strong_numbers}")

```

Output :

```

Strong numbers between 1 and 1000 are: [1, 2, 145]
PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aiaac/assign_6(8).py"
Enter the start of the range: 1
Enter the end of the range: 1000
Strong numbers between 1 and 1000 are: [1, 2, 145]
Enter the start of the range: 1
Enter the end of the range: 10000
Enter the end of the range: 1000
Strong numbers between 1 and 1000 are: [1, 2, 145]

```

Explanation :

The optimization provides dramatic speedup for large ranges. Without memoization, a range like 1-10,000 recalculates 5! thousands of times. With memoization, each factorial computes only once, delivering consistent $O(1)$ lookup for subsequent calls.

Task #9 – Few-Shot Prompting for Nested Dictionary Extraction Objective

Use few-shot prompting (2–3 examples) to instruct the AI to create a function that parses a nested dictionary representing student information.

Requirements

- The function should extract and return:

- o Full Name

- o Branch

- o SGPA

Expected Output

A reusable Python function that correctly navigates and extracts values from nested dictionaries based on the provided examples **Code**

:

```

C:\Users\PC> cd Downloads & cd aiaac & python assign_6(9).py > ...
1 """
2 Ravi Kumar, CSE, 8.7
3 Anita Sharma, ECE, 9.1
4 Suresh Reddy, ME, 7.9
5
6 """
7 def parse_student_data(data):
8     students = []
9     for line in data.strip().split('\n'):
10         name, dept, gpa = line.split(',')
11         student = {
12             'name': name,
13             'department': dept,
14             'gpa': float(gpa)
15         }
16     return students
17
18 if __name__ == '__main__':
19     data = """Ravi Kumar, CSE, 8.7
20 Anita Sharma, ECE, 9.1
21 Suresh Reddy, ME, 7.9"""
22     student_list = parse_student_data(data)
23     for student in student_list:
24         print(student)

```

Output :

```

PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aiaac/assign_6(9).py"
{'name': 'Ravi Kumar', 'department': 'CSE', 'gpa': 8.7}
{'name': 'Anita Sharma', 'department': 'ECE', 'gpa': 9.1}
{'name': 'Suresh Reddy', 'department': 'ME', 'gpa': 7.9}

```

Task Description #10 (Loops – Perfect Numbers in a Range)

Task: Generate a function using AI that displays all Perfect Numbers within a user-specified range (e.g., 1 to 1000).

Instructions:

- A Perfect Number is a positive integer equal to the sum of its proper divisors (excluding itself). o Example: $6 = 1 + 2 + 3$, $28 = 1 + 2 + 4 + 7 + 14$.
- Use a for loop to find divisors of each number in the range.
- Validate correctness with known Perfect Numbers (6, 28, 496...).
- Ask AI to regenerate an optimized version (using divisor check only up to $\sqrt{n}$). **Code :**

```

C: > Users > PC > Downloads > aiac > assign_6(10).py > ...
1  # Generate a python code to display all perfect numbers within a given range using loops
2  def is_perfect_number(num):
3      if num < 2:
4          return False
5      divisors_sum = sum(i for i in range(1, num) if num % i == 0)
6      return divisors_sum == num
7  def perfect_numbers_in_range(start, end):
8      perfect_numbers = []
9      for num in range(start, end + 1):
10         if is_perfect_number(num):
11             perfect_numbers.append(num)
12     return perfect_numbers
13 if __name__ == "__main__":
14     start_range = int(input("Enter the start of the range: "))
15     end_range = int(input("Enter the end of the range: "))
16     perfect_numbers = perfect_numbers_in_range(start_range, end_range)
17     print(f"Perfect numbers between {start_range} and {end_range}: {perfect_numbers}")
18

```

```

# optimise the above code using divisor check only up to sqrt(n)
import math
def is_perfect_number_optimized(num):
    if num < 2:
        return False
    divisors_sum = 1 # 1 is a divisor of all numbers
    for i in range(2, int(math.sqrt(num)) + 1):
        if num % i == 0:
            divisors_sum += i
            if i != num // i:
                divisors_sum += num // i
    return divisors_sum == num
def perfect_numbers_in_range_optimized(start, end):
    perfect_numbers = []
    for num in range(start, end + 1):
        if is_perfect_number_optimized(num):
            perfect_numbers.append(num)
    return perfect_numbers
if __name__ == "__main__":
    start_range = int(input("Enter the start of the range: "))
    end_range = int(input("Enter the end of the range: "))
    perfect_numbers = perfect_numbers_in_range_optimized(start_range, end_range)
    print(f"Perfect numbers between {start_range} and {end_range}: {perfect_numbers}")

```

Output :

```

PS C:\Users\PC> & C:/Users/PC/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/PC/Downloads/aiac/assign_6(10).py"
Enter the start of the range: 1
Enter the end of the range: 1000
Perfect numbers between 1 and 1000: [6, 28, 496]
Enter the start of the range: 1
Enter the end of the range: 1000
Perfect numbers between 1 and 1000: [6, 28, 496]

```